From WS–PS to the Phillips Curve

BS2591 (Carlin & Soskice 2024, Ch. 2, eqs. 2.3–2.4, Figs. 2.17–2.18)

0. Notation and conventions

- Markups: $\mu \equiv \dfrac{P - MC}{P}$ (on price), $\quad \mu^C \equiv \dfrac{P - MC}{MC}$ (on cost), $\quad \dfrac{1}{1+\mu^C} = 1 - \mu$.
- Growth rates use last period as the base: $\left(\dfrac{\Delta x}{x}\right)_t \equiv \dfrac{x_t - x_{t-1}}{x_{t-1}}$; $\quad$ inflation $\pi_t \equiv \left(\dfrac{\Delta P}{P}\right)_t = \dfrac{P_t - P_{t-1}}{P_{t-1}}$.
- $w = W/P$ is the real wage; $\lambda = y/N$ is labour productivity (constant).

1. The two curves

Price setting (PS). Firms set $P = (1 + \mu^C)\, W/\lambda$, which pins down the real wage:

$$w^{PS} = \frac{W}{P} = \frac{\lambda}{1+\mu^C} = \lambda(1-\mu) \qquad \Big(\text{general form: } w^{PS} = \lambda F(\mu, z_p)\Big).$$

With λ and μ constant, PS is horizontal.

Wage setting (WS). In general $w^{WS} = B(N, z_w)$, increasing in employment. For the Phillips curve the textbook uses the linear form in terms of the output gap:

$$w^{WS}(y_t) = B + \alpha\,(y_t - y_e) + z_w,$$

where now B is a *constant* (unemployment benefit / utility of being unemployed).

2. The bargaining gap

Suppose the economy is at the medium-run equilibrium in period -1: $y_{-1} = y_e$ and WS crosses PS, so

$$w_{-1} = w_e = B + \alpha(y_e - y_e) + z_w = B + z_w \;\; (= w^{PS}).$$

Price setting always brings the real wage back to w_e, so forward-looking wage setters try to raise the real wage above it by

$$w^{WS}(y_t) - w_{-1} = \big(B + \alpha(y_t - y_e) + z_w\big) - (B + z_w) = \underbrace{\alpha(y_t - y_e)}_{\text{bargaining gap}}. \qquad ①$$

3. Wage inflation

Under adaptive expectations wage setters expect prices to rise by last period's inflation, $\pi_t^E = \pi_{t-1} = (\Delta P/P)_{t-1}$. Since *growth of real wage* $\approx$ *growth of nominal wage* $-$ *growth of prices*, the real-wage increase they aim for is

$$\left(\frac{\Delta W}{W}\right)_t - \left(\frac{\Delta P}{P}\right)_{t-1} \;\approx\; w^{WS}(y_t) - w_{-1}. \qquad ②$$

Combining ① and ②:

$$\boxed{\left(\frac{\Delta W}{W}\right)_t \approx \left(\frac{\Delta P}{P}\right)_{t-1} + \alpha(y_t - y_e)} \qquad \text{(wage inflation)} \qquad ③$$

Wage setters cover last period's inflation *plus* the bargaining gap.

4. Price inflation

Firms set prices right after the wage round, keeping μ^C constant: $P_t = (1+\mu^C)\, W_t/\lambda_t$. Taking logs and differencing over time,

$$\log P_t = \log(1+\mu^C) + \log W_t - \log \lambda_t \implies \boxed{\left(\frac{\Delta P}{P}\right)_t = \left(\frac{\Delta W}{W}\right)_t - \left(\frac{\Delta \lambda}{\lambda}\right)_t} \qquad \text{(price inflation)} \ ④$$

5. The Phillips curve

With constant productivity, $(\Delta\lambda/\lambda)_t = 0$, so ④ gives $(\Delta P/P)_t = (\Delta W/W)_t$: firms pass wage increases one-for-one into prices. Substituting ③:

$$\left(\frac{\Delta P}{P}\right)_t = \left(\frac{\Delta P}{P}\right)_{t-1} + \alpha(y_t - y_e). \qquad ⑤$$

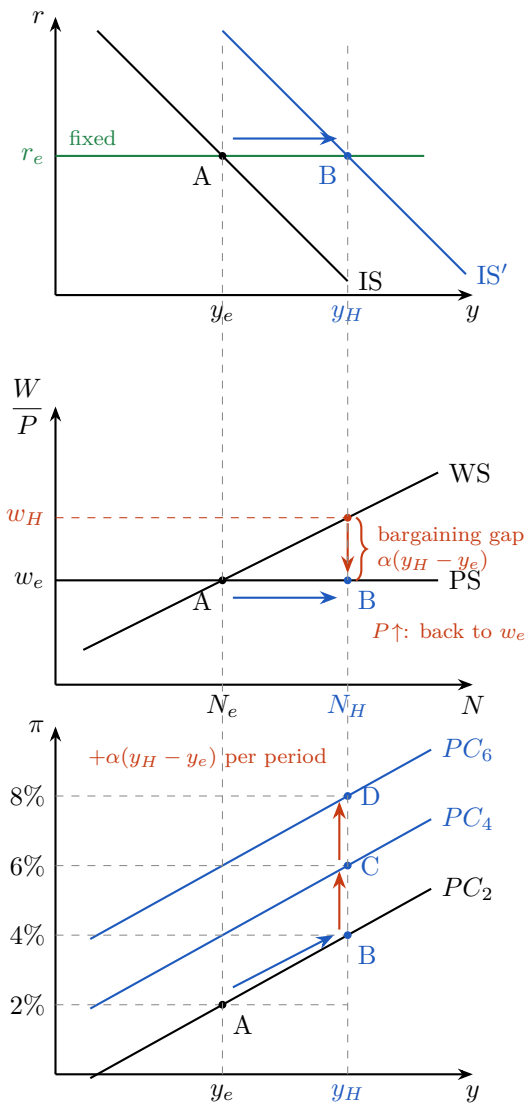
Writing $\pi_t \equiv (\Delta P/P)_t$:

$$\boxed{\underbrace{\pi_t}_{\text{current inflation}} = \underbrace{\pi_{t-1}}_{\text{lagged inflation}} + \alpha \overbrace{(y_t - y_e)}^{\text{output gap}}_{\text{bargaining gap}}} \qquad \text{(PC, eq. 2.4)}$$

- **Height:** at $y_t = y_e$, $\pi_t = \pi_{t-1} = \pi_t^E$. We label each PC by expected inflation, e.g. $PC(\pi^E = 2\%)$, or PC_2.
- **Slope:** α, the slope of WS. A steeper WS means a steeper PC.

6. Shock (a): a positive aggregate demand shock

Colour key: **Assumption** (held fixed) **Change** (what moves) **Mechanism** (why it moves)



Step 0 (start, period −1). Economy at A: $y = y_e$, $\pi = 2\%$, $w = w_e$. The central bank keeps r at r_e (no policy response yet; that comes in Ch. 3). WS, PS, λ, μ^C do not change, so y_e does not change.

Step 1: demand shock. Investment boom: IS shifts right to IS′. At r_e output rises $y_e \to y_H$ and employment $N_e \to N_H$ (A→B in the top two panels). Because r stays at r_e, the whole demand increase shows up in output.

Step 2: wage round. At N_H the WS curve lies above PS: wage setters now want $w_H > w_e$. Lower unemployment strengthens workers' bargaining position. The gap $w_H - w_e = \alpha(y_H - y_e)$ (say 2%) is the bargaining gap, so $\Delta W/W = 2\% + 2\% = 4\%$ by ③.

Step 3: price setting. Prices rise 4%: inflation goes $2\% \to 4\%$, a move *along* PC_2 from A to B. Firms keep their markup, so by ④ they pass the 4% cost increase into prices. The real wage is pushed back to w_e on PS; workers do not get w_H.

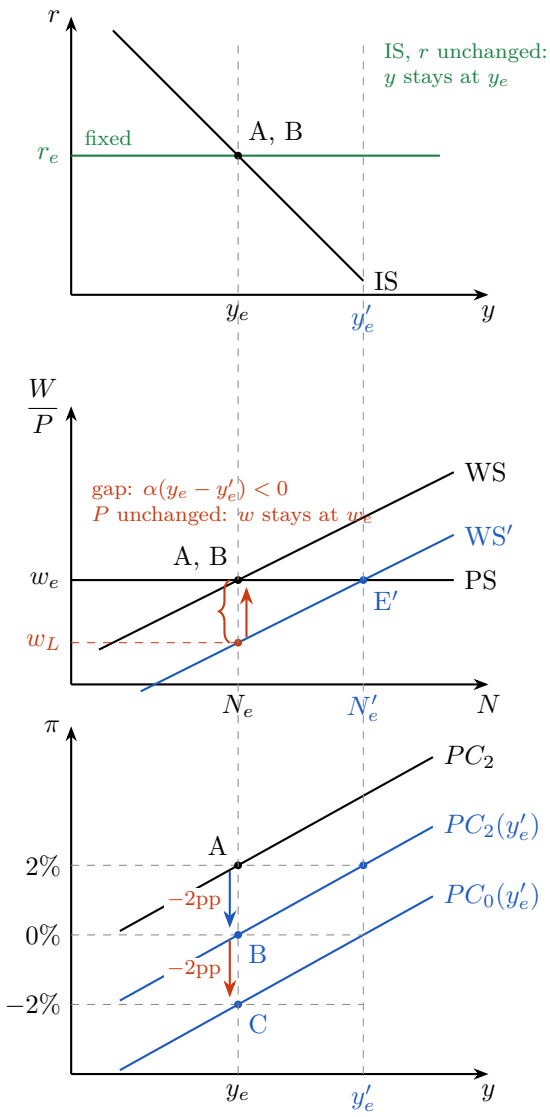
Step 4: next periods. $PC_2 \to PC_4 \to PC_6$; inflation $4\% \to 6\% \to 8\%$ (B→C→D). Output stays at y_H because r stays at r_e, so the output gap persists. The real wage is pinned at w_e, so the same bargaining gap reopens at every wage round. With $\pi^E_t = \pi_{t-1}$ now higher, each PC sits $\alpha(y_H - y_e)$ above the last one.

Result. As long as $y > y_e$, inflation keeps rising (*accelerationist* PC; "conflict inflation"). It stops rising only when output returns to y_e.

Period	Output	Lagged π	Bargaining gap	$\Delta W/W$	π	Real wage
−1	y_e	2%	0	2%	2%	w_e
0	y_H	2%	2%	4%	4%	w_e
1	y_H	4%	2%	6%	6%	w_e
2	y_H	6%	2%	8%	8%	w_e

7. Shock (b): a positive aggregate supply shock

Same colour key as Section 6. The example in the textbook is a fall in union bargaining power (e.g. the Dutch reforms of the 1980s: less generous unemployment benefits, active labour market policies, more coordinated wage bargaining).



Step 0 (start, period −1). Economy at A: $y = y_e$, $\pi = 2\%$, $w = w_e$. No stabilising policy maker: r stays at r_e and IS does not move. PS, λ and μ^C are unchanged.

Step 1: supply shock. Weaker unions shift WS down to WS′ (for an upward-sloping WS, down = right). The new WS′–PS intersection raises equilibrium employment $N_e \to N'_e$ and output $y_e \to y'_e$. The equilibrium real wage is still w_e, because PS has not moved. A higher cost of job loss means workers supply effort at a lower real wage at every N.

Step 2: output does not move. y stays at y_e (A and B coincide in the top panel), so now $y < y'_e$. Output is set by demand. With IS and r unchanged, nothing raises demand. The output gap is measured from the *new* equilibrium: $y_e - y'_e < 0$.

Step 3: wage round. At N_e, WS′ lies below PS: wage setters accept $w_L < w_e$. Nominal wages rise by $\Delta W/W = 2\% - 2\% = 0\%$ (eq. 2.5). The bargaining gap is negative: $\alpha(y_e - y'_e) = -2\%$, so it cancels lagged inflation.

Step 4: price setting. Prices are unchanged: inflation falls $2\% \to 0\%$ (A→B). The PC shifts down to $PC_2(y'_e)$, which is anchored at the new y'_e. Costs did not rise and the markup is constant, so $\pi = 0\% - 0\% = 0\%$ (eq. 2.6). The real wage stays at w_e.

Step 5: next periods. Inflation keeps falling, $0\% \to -2\% \to -4\%$ (B→C→ ⋯): deflation. The PC keeps shifting down ($PC_0(y'_e)$, …). Output stays at $y_e < y'_e$, so the negative gap reopens every wage round, and $\pi^E_t = \pi_{t-1}$ is lower each time.

Result. A positive supply shock means falling inflation at the old output level, for as long as $y < y'_e$. The economy *could* run at y'_e with constant inflation; in Ch. 3 the central bank cuts r to get there.

Period	Output	Lagged π	Bargaining gap	$\Delta W/W$	π	Real wage
−1	y_e	2%	0	2%	2%	w_e
0	y_e $(< y'_e)$	2%	−2%	0%	0%	w_e
1	y_e $(< y'_e)$	0%	−2%	−2%	−2%	w_e
2	y_e $(< y'_e)$	−2%	−2%	−4%	−4%	w_e

8. Comparing the two shocks (r fixed at r_e)

	Demand shock (IS shifts)	Supply shock (WS shifts)
Output y	changes: $y_e \to y_H$	unchanged at y_e
Equilibrium output y_e	unchanged	changes: $y_e \to y_e'$
Output gap	$y_H - y_e > 0$	$y_e - y_e' < 0$
On the PC diagram	move *along* PC_2, then PC shifts up each period	PC re-anchored at y_e' (shifts down), then keeps shifting down
Inflation	rises every period	falls every period